Program : **Diploma in Automation and Robotics**	
Course Code : **6338**	Course Title: **HDL and Simulation Software Lab**
Semester : **6**	Credits: **1.5**
Course Category: **Program Elective**	
Periods per week: **3 (L:0, T:0, P:3)**	Periods per semester: **45**

Course Objectives:

- Students will be able to take up and design advanced digital systems using FPGA

General Instructions:

- Students have to write the date, aim, Software & Hardware requirements for that experiment in the observation book.
- Students have to listen and understand the experiment explained by the faculty and note down the important points in the observation book.
- Students need to write procedure/algorithm in the observation book.
- Analyze and Develop/implement the logic of the program by the student in respective platform
- After approval of logic of the experiment by the faculty then the experiment has to be executed on the system.
- After successful execution the results are to be shown to the faculty and noted the same in the observation book.
- Students need to attend the Viva-Voce on that experiment and write the same in the observation book.
- Update the completed experiment in the record and submit to the concerned faculty in-charge.

Course Prerequisites:

Topic	Course code	Course name	Semester
Analog and Digital Electronics	3331	Analog and Digital Electronics	3

Tools and Equipment:

Processors: Intel Atom® processor or Intel® Core™ i3 processor.
Disk space: 1 GB.
Operating systems: Windows 7 or later, macOS, and Linux.
Software : HDL Simulator

Course Outcomes:

On completion of the course, the student will be able to:

Con	Description	Duration (Hours)	Cognitive level
CO1	To familiarize students with HDL simulation	24	Applying
CO2	To implement complex digital systems using FPGA.	18	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		
CO2	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**To familiarize students with HDL simulation**		
M1.01	Write a program in prolog to implement simple facts and Queries	4	Applying
M1.02	Write a program in prolog to implement simple arithmetic	4	Applying
CO2	**To implement complex digital systems using FPGA.**		

M2.01	Simulate HDL program for the implementation of Adder circuit.	2	Applying
M2.02	Simulate HDL program for the implementation of Flip flops.	2	Applying
M2.03	Simulate HDL program for the implementation of Counters	2	Applying
M2.04	Simulate HDL program for the implementation of Comparators	2	Applying
M2.05	Simulate HDL program for the implementation of MUX	2	
M2.06	Simulate HDL program for the implementation of Encoder circuits.	2	
M2.07	Simulate HDL program for the implementation of Shift Registers	2	
M2.08	Simulate HDL program for the implementation of FSM	2	
M2.09	Simulate HDL program for the implementation of RAM	2	
M2.10	Simulate HDL program for the implementation of PWM	2	
	Lab Exam I	1.5	
CO3	**To implement complex digital systems using FPGA**		
M3.01	To implement ADC and DAC using FPGA	6	Applying
M3.02	To implement UART Transceiver using FPGA	4	Applying
M3.03	To implement VGA monitor using FPGA	4	Applying
M3.04	To implement Ethernet using FPGA	4	Applying
	Lab Exam II	1.5	

Online Resources:

Sl.No	Website Link/Textbooks
1	Ott, Douglas E. and Thomas J. Wilderotter. A Designer's Guide to VHDL Synthesis. New York, NY: Springer, 1994. ISBN: 0792394720.
2	Fletcher, W. I. An Engineering Approach to Digital Design. Englewood Cliffs, NJ: Prentice-Hall, 1980. ISBN: 0132776995.
3	Mano, M. Morris. Computer Engineering: Hardware Design. Englewood Cliffs, NJ: Prentice-Hall, 1988. ISBN: 0131629263.
4	https://github.com/